

Calculator Questions

Quadratic Equations

Solving Quadratics by Factorising / The Quadratic Formula / Solving Quadratics using a Calculator

Medium (7 questions)	/21
Hard (4 questions)	/13
Very Hard (3 questions)	/13
Total Marks	/47

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Medium Questions

- 1 Solve $6x^2 - 7x + 2 = 0$.

You must show all your working.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(3 marks)

- 2 Use the quadratic formula to solve $9x^2 - 12x - 23 = 0$.

Give your answers correct to 2 decimal places.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(3 marks)

- 3 Solve the equation $3x^2 - 2x - 10 = 0$.

Show all your working and give your answers correct to 2 decimal places.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(3 marks)

- 4 Use the quadratic formula to solve the equation $3x^2 + 7x - 11 = 0$.

You must show all your working and give your answers correct to 2 decimal places.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(3 marks)

- 5 Solve the equation $3x^2 - 2x - 2 = 0$.

Show all your working and give your answers correct to 2 decimal places.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(3 marks)

- 6 Solve the equation $6m^2 + 25m + 16 = 0$.

Show all your working and give your answers correct to 2 decimal places.

$m = \dots\dots\dots$ or $m = \dots\dots\dots$

(3 marks)

- 7 Solve the equation $2x^2 + 7x - 3 = 0$.

Show all your working and give your answers correct to 2 decimal places.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(3 marks)

Hard Questions

- 1 (a)** Nina runs 5 km at y km/h and then walks 7 km at $(y - 7)$ km/h.
She takes a total of 2 hours.

Show that $2y^2 - 26y + 35 = 0$.

(3 marks)

- (b)** Solve $2y^2 - 26y + 35 = 0$.

$y = \dots\dots\dots$ or $y = \dots\dots\dots$

(3 marks)

- (c)** Find Nina's walking speed

$\dots\dots\dots$ km/h

(1 mark)

- 2** Solve by factorisation.

$$3x^2 - 11x + 6 = 0$$

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(2 marks)

- 3** The n th term of a sequence is $4n^2 + n + 3$.
Find the value of n when the n th term is 498.

$n = \dots\dots\dots$

(2 marks)

- 4** Solve by factorisation $10m^2 - 31m - 14 = 0$.

$m = \dots\dots\dots$ or $m = \dots\dots\dots$

(2 marks)

Very Hard Questions

- 1 Find all the solutions to the equation.

$$(x^2 - 4x + 4)(x^2 - 9x + 20) = 1$$

(4 marks)

- 2 (a) Solve $2x^2 + 9x - 7 = 0$

You must show all your working and give your solutions correct to 3 significant figures.

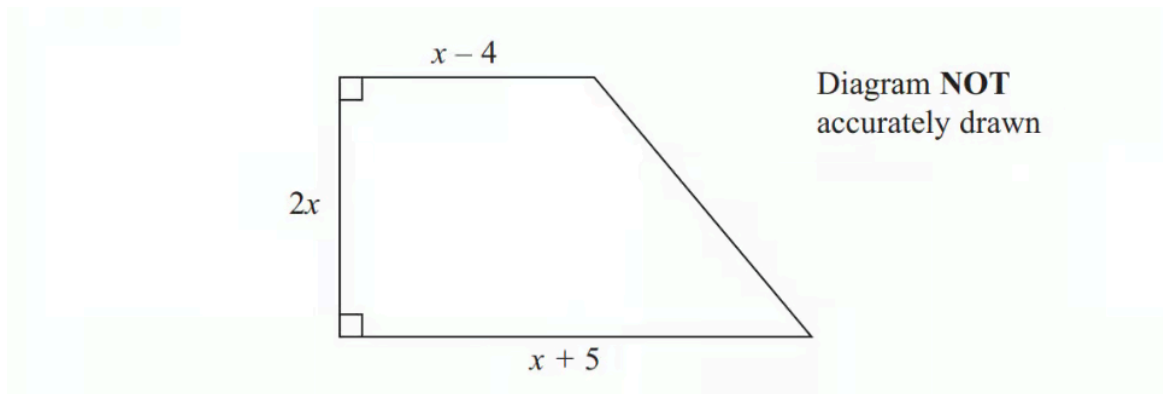
(3 marks)

- (b) Use your answer to part (a) to solve $\frac{2}{y^2} + \frac{9}{y} - 7 = 0$

Give your solutions correct to 3 significant figures.

(2 marks)

3 (a) The diagram shows a trapezium.



All the measurements are in centimetres.

The area of the trapezium is 351 cm^2 .

Show that $2x^2 + x - 351 = 0$

(2 marks)

(b) Work out the value of x .

(2 marks)